

Blockchain-Based AI Model Integrity and Verification Framework

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Abstract. This paper presents a novel framework for ensuring the integrity and verifiability of AI models using blockchain technology, specifically leveraging the Solana blockchain's off-chain capabilities. As artificial intelligence continues to play an increasingly critical role across various sectors, the need for transparent, traceable, and auditable AI model management has become paramount. Our proposed solution addresses this challenge by implementing a tokenization approach for AI models, which enables secure tracking of model lineage, modifications, and ownership transfers.

The framework utilizes Solana's high-performance blockchain and the Token-2022 program to create a tamper-resistant record of AI model provenance and metadata. By incorporating parent-child relationships between tokens, the system facilitates the tracing of model evolution and specialization. This approach enhances trust in AI systems by providing a verifiable model development and deployment history.

We discuss the implementation details, including the off-chain solution architecture, tokenization process, and metadata structure. The paper also explores potential future developments, such as on-chain implementations, advanced encryption techniques, and creating an AI model marketplace. Our work contributes to the growing field of blockchain applications in AI governance, offering a scalable and efficient solution for maintaining AI model integrity in an increasingly complex technological landscape.

Keywords: Blockchain · AI Model Integrity · Solana · Tokenization · Model Verification

1 Introduction

In the previous several years since the public introduction of ChatGPT [1] the concept of artificial intelligence has become more prominent in public. Generative AI, exemplified by models like GPT-3 [2] and DALL-E [3] has demonstrated remarkable capabilities in natural language processing, image generation, and creative tasks such as problem-solving [4] or coding assistance [5]. According to a report by McKinsey, generative AI could add between $2.6 trillion to $4.4 trillion annually to the global economy [6].

The rise of AI models has been prominent in industry and academia. In the field of natural language processing, the introduction of the Transformer architecture [7] it

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marked a significant turning point, leading to the development of increasingly powerful language models. Besides language models, exemplified by Large Language Models (LLMs), AI models encompass any set of given parameters that, when applied to a specific artificial intelligence architecture (such as neural networks) provided an expected set of outputs determined by training [8]. An example would be the YOLO model [9] used for real-time object detection or ResNet-50 [10] used for image classification.

A survey by PwC [11] found that companies believe AI, most notably Generative AI, will provide them with a "significant edge" in the future. The same report shows that AI will have its "truth moment" soon, where additional controls should occur to prevent AI hallucinations and copyrighted content, for which the authors have already investigated the leading causes inside the ChatGPT platform [12].

In the 12th Annual Survey of Fortune 1000 and Global Data and AI Leadership [13] which interviewed C-level executives regarding data and AI. One of the significant findings is that companies will continue investing in data and analytics. They also believe that data and AI safeguards are necessary since only 15.9% of participants say there are enough safeguards for data and AI ethics. Furthermore, they think data and AI should be integrated into traditional business processes.

The rise of AI models represents a technological shift powered by previous knowledge and experiences, with the potential for further business and societal transformation. Due to this paradigm shift, the need arises for the ability to manage AI models, their distribution, and how they are connected to provide an ethical and efficient approach for the future.

2 Problem Statement

The rapid advancement and widespread adoption of **artificial intelligence** (AI), particularly in generative AI, have led to unprecedented challenges in ensuring the integrity, transparency, and verifiability of all kinds of AI models, not only generative ones. As these models become increasingly complex and influential in decision-making processes across various sectors, the need for a robust framework to maintain their integrity and facilitate verification has become critical.

One of the first problems we have encountered is that there is no standardized, tamper-proof system for tracking the origin, development history, and modifications of AI models. This absence makes it difficult to establish the authenticity and lineage of models, potentially undermining trust in AI systems. Systems such as HuggingFace exist, but with problems of their own. One of the common problems is the centralization of models and censorship. Another issue with these existing issues is the lack of model signing by the authors regardless of the platform. Without additional model signatures or similar checks, a malicious attacker can exploit them for supply chain attacks [14].

Traditional version control systems are often insufficient for managing the complexities of large-scale AI models, including their numerous iterations, fine-tuned versions, and the vast datasets used in their training. This leads to potential abuse of AI models and a lack of governance between different variants of the AI models. Unless specific AI/ML platforms are used, such as Weights & Biases, versioning can become problematic, especially in the case of multiple variants of a single model (such as GPT-3 model

variants with different context sizes or numbers of tokens used for training [2]). With the cost of training being expensive [15], it is necessary to track model versions to lower costs.

Without model verification, unauthorized alterations could introduce biases, backdoors, or other potentially problematic vulnerabilities. Similarly, the lack of audibility makes it difficult to audit their decision-making by assessing the training data and hyperparameters utilized. This lack of transparency is particularly problematic in sensitive applications such as healthcare or finance, where there is a direct correlation with high-risk decision-making. Implementing advanced features of data management, such as read-only data, access control and timestamps would help this issue, depending on the type of data stored and the solution configuration, as well as the type of the data store.

The ecosystem lacks a standardized framework for securely sharing and collaboratively developing AI models across organizations or research institutions. Solutions exist, as the beforehand mentioned HuggingFace or Weights and Biases [16], but they lack compatibility layers and ways for models to be interchangeable depending on the platform or model comparison between platforms which would answer the question which model being used is the real one. Other platforms exist such as Amazon SageMaker [17] or Azure Machine Learning [18] which are tightly coupled with their parent ecosystems (Amazon and Azure), making it difficult to share models between different platforms.

An alternative approach to AI model distribution and security involves building custom solutions using proven technologies like Kubernetes and Docker. While this approach gives teams more control over their systems and the ability to customize features based on specific needs, it comes with important trade-offs to consider. Creating such a system requires significant development time and technical expertise to build something stable enough for production use. These custom solutions can be complex to manage, often running into practical challenges with keeping software versions compatible across different parts of the system, as well as complexity issues from the number of components being orchestrated.

These interconnected problems underscore the need for a comprehensive solution to address AI model integrity and verification challenges. Such a solution seems suited for applying blockchain technology—including immutability, decentralization, and transparency—to create an ecosystem for AI model management. Addressing these issues, the proposed solution aims to enhance trust in AI models and their implementation and provide a mechanism for traceability and accountability between different AI model variants.

3 Current Work

AI model distribution has seen significant advancements in recent years, with a growing focus on leveraging blockchain technology to address model integrity, transparency, and security challenges. This analysis synthesizes insights from crucial sources, exploring general AI model distribution techniques and blockchain-specific applications in this domain.

There is a demand for frameworks that handle AI model distribution [19] as well as systems that handle additional security of AI models to prevent tampering [20] by using additional deep learning and compression. As the AI model landscape increases, these systems will prove invaluable in the foreseeable future.

Blockchain has already been used for the concept of sharing AI models [21] with the concept of tokenization, but this approach primarily covers the legal side of the AI model tokenization. Such research points to the fact that AI model tokenization is a possible approach for AI model distribution.

Blockchain is also the solution of choice for handling legal contracts [22] due to the immutability of the data stored. If we abstract the concept, we can see that blockchain is a great candidate as an integrity store of sensitive data that is sensitive to modifications and tampering.

Besides the technical implementation of blockchain and AI models, there is also research about the development and need for AI model marketplaces [23]. The need for these marketplaces stems from the increasing prominence of AI models in the industry.

Current blockchain solutions for AI model management are primarily dominated by specialized platforms like Ocean Protocol [24] and SingularityNET [25], which were initially developed on the Ethereum blockchain. These platforms have created comprehensive ecosystems for AI model governance, trading, and verification, but their underlying blockchain architecture presents certain limitations. Ocean Protocol, which remains on Ethereum, leverages its OCEAN token for facilitating transactions and access control in its AI and data marketplace. The platform's strength lies in its sophisticated compute-to-data functionality and privacy-preserving features, allowing models to be utilized without exposing sensitive data. However, its reliance on Ethereum means dealing with relatively high gas fees and limited transaction throughput of approximately 15 transactions per second [26].

SingularityNET, originally built on Ethereum but now migrated Cardano, uses its AGIX token for platform operations. This migration was driven by the need for better scalability and lower transaction costs, highlighting the limitations of the Ethereum network for AI model management at scale. The platform offers unique capabilities in AI service composition and marketplace functionality but still operates within the constraints of its underlying blockchain architecture.

These two platforms have expressed interest into merging to maintain their business model and improve blockchain capabilities in the domain of AI model management [27].

With the research performed so far, we can confirm that there is not an efficient and complete blockchain-based solution for AI model distribution, verification, and integrity management but that such solutions are needed now and for the future.

4 Blockchain as an Integrity Store

An integrity store is a system or mechanism designed to maintain and verify the authenticity, consistency, and trustworthiness of data or digital assets over time [28]. The primary functions of an integrity store include data immutability, verifiability, auditability, and tamper resistance.

Data immutability is defined as the intrinsic capability of data that cannot be changed after publishing (in contrast to mutability, where data can be changed or mutated after publishing) [29], while verifiability provides a mechanism for the data to be authenticated and verified. These two points relate to tamper-resistance, which handles the ability of stored data to be resilient to potential changes.

Blockchain technology embodies the characteristics of an ideal integrity store, making it particularly well-suited for maintaining the integrity of AI models and other digital assets. Blockchain, by definition, provides immutability [30] due to cryptography used for signing data. Blockchain data is stored in hashed blocks and then connected with previous blocks, creating a chain of hashes. These chains can take different shapes, most notably in the form of a list, such as Bitcoin, or the shape of Direct Acyclic Graphs (DAGs), such as Solana [31].

Verifiability is an additional property provided by immutability. It is implemented by design since it can be verified using the identical cryptographic hashes used to confirm it.

Due to the linkage between different blocks and their cryptographic signing, performing a data audit across the blockchain is possible. This gives us the entire history and documents the data change process as it gets published onto the blockchain.

Blockchain solutions are distributed by design and organized by consensus algorithms. Consensus algorithms represent a family of algorithms that organize and synchronize states between distributed systems. Consensus algorithms guarantee consistency between data distributed among blockchain nodes. An example of a consensus algorithms is the Proof-of-Work algorithm [30] used by Bitcoin or the Proof-of-History algorithm used by Solana [31].

The application of blockchain technology emerges as an excellent choice for AI model distribution thanks to its features that align with the requirements of an integrity store. Its ability to ensure data immutability, verifiability, and tamper resistance makes it an ideal platform for maintaining the integrity of AI models throughout their lifecycle. By leveraging blockchain's cryptographic foundations and auditable structure, AI model developers can ensure the authenticity and consistency of their models.

5 Solution Implementation Details

5.1 Solution Architecture and Critical Points

We propose a novel solution leveraging the Solana blockchain infrastructure to address the challenges of distributed version control and data integrity in modern systems. This approach capitalizes on blockchain's inherent technological advantages to provide a robust, scalable versioning system. The fundamental blockchain architecture provides built-in versioning capabilities through its inherent block structure, where each block maintains a cryptographic link to its parent block, creating an immutable chain of relationships that naturally supports version control mechanisms. The system employs two primary mechanisms for ensuring data integrity: cryptographic hashing of data within blocks and parent-child relationship validation between consecutive blocks. The distributed nature of blockchain systems provides significant advantages in terms of fault

tolerance through distributed node architecture, automated rollback capabilities to verified states, and system-wide consistency maintenance. Data verification in the system relies on a consensus algorithm that ensures agreement across distributed nodes, maintains data integrity throughout the network, and validates state changes across the system.

When compared to centralized versioning platforms such as HuggingFace or Weights & Biases, our solution offers distinct advantages through its distributed architecture, which eliminates single points of failure, provides cryptographic verification of all stored data, enables precise timestamping of all changes, and ensures verifiable linking to historical data states. The system supports two primary methods for expressing relationships between data blocks: direct linkage through traditional parent-child relationships between consecutive blocks, and graph-based relationships enabling advanced relationship modeling. The latter allows for complex dependency mapping, multi-dimensional version tracking, and detailed relationship expression between non-consecutive blocks.

Solana [31] is a high-performance blockchain platform based on **Direct Acyclic Graphs** (DAGs) that was created in 2018 by Anatoly Yakovenko. It aims to address the scalability issues faced by many existing blockchain networks. Solana's development was driven by the need for a fast, secure, and scalable blockchain that could support decentralized applications (dApps) and crypto-economic systems. The project gained significant attention and adoption in 2021 during the boom in decentralized finance (DeFi) [32] and non-fungible tokens (NFTs) [33].

From a technological standpoint, Solana introduces several new features. Its consensus is performed by the proof-of-history (PoH) consensus algorithm, which works alongside a proof-of-stake (PoS) based algorithm. PoH creates a historical record proving that an event has occurred at a specific time, allowing for faster transaction processing. This enables Solana to achieve high throughput with low transaction costs. These technological advancements position Solana as a potential solution for scalable blockchain applications. Compared to existing blockchain solutions – Ocean Protocol and SingularityNET, Solana provides better transaction throughput, lower transaction costs and better scalability, as well as a stricter environment for program development due to usage of Rust.

Our solution's current state is based on the off-chain implementation, which works with the Solana blockchain. Another option for further development is to use a Solana on-chain program to handle the proposed solution.

Off-chain solutions benefit from easier usage than on-chain programs, integration with external systems, and extensibility with common programming languages, depending on the needs. Since Solana programs that run on-chain are written in Rust, which has a specific learning curve, they can be used by any programming language.

Performance wise, off-chain solutions based on Solana utilize **Zero Knowledge** (ZK) rollups and optimistic rollups to increase overall performance and throughput. ZK rollups [34] leverage cryptographic validity proofs to verify transaction authenticity, enabling immediate finality once proofs are verified on-chain, albeit at the cost of higher computational complexity in proof generation. This mathematical approach employs zero-knowledge proof systems [35], to compress transaction data while maintaining verifiable integrity. In contrast, optimistic rollups [36] operate under the assumption

of transaction validity, implementing a challenge period during which validators can contest potentially fraudulent transactions through fraud proofs. While this methodology significantly reduces computational overhead during normal operation, it introduces a time delay for transaction finality. The overall choice between ZK rollups or optimistic rollups lies in the trade-off between computation performance needed to confirm the off-chain transactions and transaction settlement periods.

One of key benefits running the solution on-chain would be in the extensibility, and the utilization of advanced Solana features such as PDA [37]. The implementation of a token factory pattern would improve security and automate the integrity and verification framework.

For the user to tokenize their AI model and its information to the Solana blockchain, the user needs to publish their AI model, as seen in Fig. 1. Publishing the AI model involves deploying the model to some arbitrary location (such as IPFS or another data store) where the model can be accessed. The returned model location is then passed to the off-chain script, which communicates with the Solana blockchain to tokenize it according to the metadata configured, shown in Fig. 2. The result of the **off-chain** script is a token address that contains the minted token (or tokens, depending on the configuration) of the published AI model, which can be used further to prove ownership or trading.

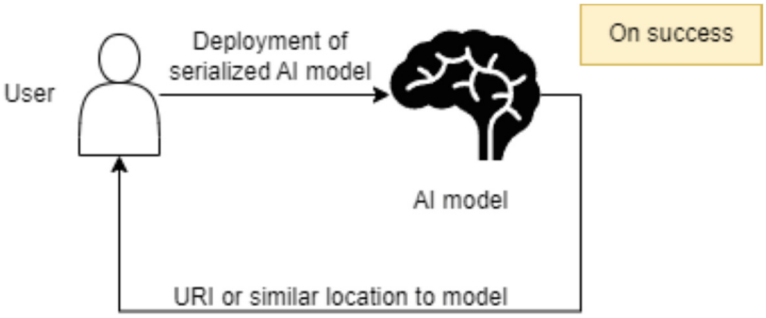


Fig. 1. AI model publishing process

This proposed solution allows AI models to contain parent information. Parent tokens can represent base AI models from which they have been derived, such as general-purpose AI models. The tokenization approach utilizes token metadata, which contains the field that allows the tokens to be linked, providing a chain of inheritance. Chain of inheritance is visualized in Fig. 3. The main benefit of this approach is the verifiability and auditability of new or specialized AI models, which helps in data management and compliance.

Besides parent data, the metadata contains additional information about the model, such as the model's name, description, short identifier, and location field, which includes the actual published AI model location.

5.2 Implementation Details

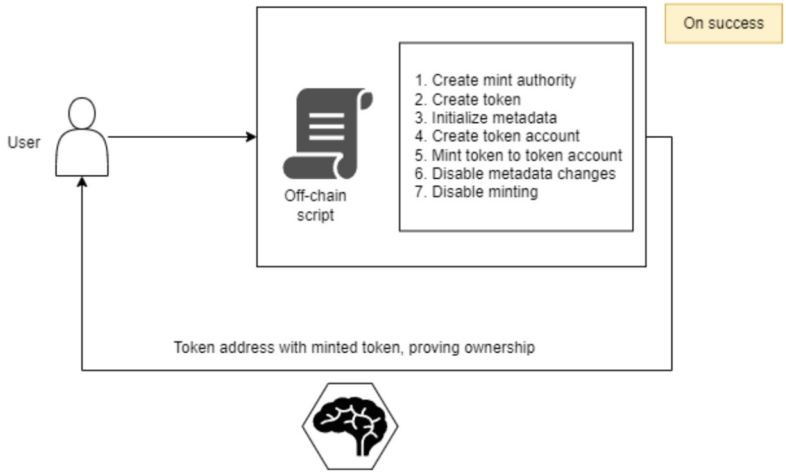


Fig. 2. AI model tokenization workflow with the off-chain Solana script

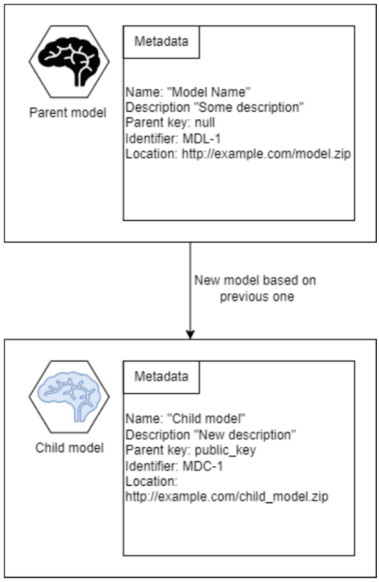


Fig. 3. Description of tokenized AI model metadata and the parent relationship between tokens

Prerequisites

The proposed solution is implemented in Python, utilizing the Solana Python RPC API. For the demonstration purposes of this paper, we will present the Python implementation by using the `Solana` and `spl-token` command line programs to abstract the entire off-chain program and show its main features. We have chosen the `spl-token` functionality to mint tokens due to its simplicity for our current use case, although further

implementations might be based on Metaplex [38], depending on the finalized use cases and features.

The authors chose Python implementation due to their familiarity with the programming language. The above commands can be implemented in any programming language which supports the **Solana RPC** API [39].

The primary prerequisite is the installation of the Solana CLI tools. These tools will be used to perform the primary unit of work for the off-chain program but also provide the `solana-validator` which we have used for development. The `solana-validator` is necessary to run a local Solana network for testing and development. In other cases, such as using the devnet or testnet, the `solana-validator` is not required.

The next step would be to create your local Solana account, which will be used as the mint authority. Mint authority represents a Solana account, which is designated and authorized to mint new tokens [40]. This account is a cryptographic key pair that allows you to store SOL tokens used by the Solana network for blockchain operations. You can create a Solana account using the `solana keygen` command. This command performs the necessary cryptographic operations to create a key-value pair, which is later used. After creation, you need to set your Solana CLI client to use the generated keypair using the `command solana config set -keypair your_public_key` where `your_public_key` represents the public key provided by the `solana keygen` command.

After token creation, if the other necessary commands are issued from a different Solana account, the operations will fail, hence centralizing the minting process to the actual owner or operator of the AI model.

If you are not using the local validator, you must obtain SOL tokens in your local account to perform the necessary tokenization activities. One of the ways to obtain SOL tokens for the devnet and testnet is by airdropping. Airdropping represents a process of acquiring blockchain tokens from an automated system [41]. In the case of Solana, airdropping via the command line is done by running the following command:

```
solana airdrop number_of_tokens
```

Where `number_of_tokens` represents the number of tokens that will be airdropped into your account.

In the case of the mainnet, you need to provide your tokens externally since airdrops are confined to the devnet and testnet.

Minting of a New Token

As mentioned above, the mint authority generates (or mints) new tokens based on the Solana network. Our solution's minting process is done with the `spl-token` command, which utilizes the Solana program library to create tokens. Our token is based on the Token-2022 program [42]. which is deployed on the following Solana address:

```
TokenzQdBNbLqP5VEhdkAS6EPFLC1PHnBqCXEpPxuEb
```

Utilizing this token program, we can optimize the token development process and increase the security of our solution because the Token-2022 program has been audited and confirmed to provide a safe implementation for token minting.

To create a new token for AI model distribution, we must make the necessary metadata containing the token information and the path to the serialized AI model.

Metadata is defined as a JSON file with the following definition:

```
{
name: string
description: string
parent_key: public_key
identifier: string
location: url
}
```

The name represents the name of the model that is being tokenized, while the description field represents the model's description. If the model has a parent (due to specialization or fine-tuning), the parent's public key (or token address) should be a part of the `parent_key` field. The identifier field represents a short code for the model, like token tickers, while the value field contains the location where the actual serialized model is stored. We have decided to use the location filed to set up as a string since the model can be deployed in various places, such as IPFS or different file storage (such as Amazon S3).

The token is minted with the following `spl-token` command:

```
spl-token create-token --program-id
TokenzQdBNbLqP5VEhdkAS6EPFLC1PHnBqCXEpPxuEb    --decimals    0
--enable-metadata
```

Where `--program-id` points to the location of the Token-2022 program, `--decimals` determine if the token can be partial, and `--enable-metadata` enables the attachment of metadata to the token. The result of this command is the token identifier, which contains the necessary token information without having it minted.

Before minting, we need to initialize the metadata to the token id and generate the token address to store the actual minted tokens. Metadata initialization is done with the following command:

```
spl-token initialize-metadata token_id token_metadata
```

Where `token_id` is the provided address from the previous step and `token_metadata` is the serialized JSON defined earlier.

Token address creation uses the `spl-token create-account token_id` command, where `token_id` is the token id defined in the previous steps.

With all the prerequisites, we can now mint the token representing our AI model. Minting is performed with the following command:

```
spl-token mint token_id token_amount
```

Where `token_id` is the token id of the defined token, and the token_amount is the number of tokens that we wish to mint. Depending on the `--decimals` field when the token was defined, the token can be single or partial. Single tokens are similar to non-fungible ones since only one instance exists. In contrast, partial tokens are based on the `--decimals` field and allow partial ownership after transferring. There is only one minting operation for the entire token lifecycle to keep the AI model unambiguous and unique across the blockchain.

Minting and metadata changes are stopped by issuing the following commands:

```
spl-token authorize token_id metadata -disable
spl-token authorize token_id mint -disable
```

At this point, the token has been created and minted, thus proving full or partial ownership of the AI model defined in the metadata, depending on the original token configuration.

Token Transfers
Transferring ownership of a tokenized AI model is the same as any other token transfer on the Solana blockchain. Token transfer can be performed with the following CLI command:

```
spl-token transfer token_address_of_sender 1
token_address_of_receiver
```

Where `token_address_of_sender` is the token address of the current token owner, while the `token_address_of_receiver` is the token address of the future owner. In this example, we assume that the token is indivisible; hence, the transfer makes a complete ownership change. Changing the value of `1` to a larger value (depending on the `--decimals` flag when the token was created) would create a partial transfer of ownership.

6 Discussion and Future Work

The proposed blockchain-based AI model integrity and verification framework represents a significant step towards enhancing transparency, accountability, and trust in the AI ecosystem. However, several areas remain for potential improvement and expansion.

While our current framework utilizes blockchain technology for model verification and integrity assurance, the tokenization process remains primarily off-chain. A fully on-chain solution for tokenization could offer advanced security measures, such as additional encryption or Solana capabilities. Besides that, automating the entire process would become possible, starting with creating token factories that would automatically generate mint accounts and verify user tokenization requests.

By moving the entire tokenization process on-chain, we can leverage the inherent security properties of blockchain technology, reducing the risk of tampering during the tokenization process.

Besides on-chain improvements, the creation of a framework similar to **the Fishbone model** [43] which could determine the universal authentication framework necessary for accessing AI models according to data sensitivity. This could be combined with the concept of digital identities inspired by e-learning [43] to keep the blockchain network users verified and actively participating, thus improving performance as the blockchain network improves.

Furthermore, on-chain tokenization would allow for real-time tracking and auditing of the tokenization process, enhancing the transparency of AI model management.

Exploring using decentralized storage networks for AI model storage with Solana could enhance data availability and resilience. The authors already covered this approach [44] by using IPFS as a distributed data store and Ethereum NFTs for model tokenization.

While our current framework provides an accountability interface and a way to automate the tedious manual process of token generation, it lacks granular access control mechanisms for model storage. Implementing additional encryption layers could significantly enhance the system's security and privacy.

Developing a blockchain-based trading marketplace for AI models represents an exciting direction for future work. It would build upon the ability to transfer tokens while extending it to rent, sell, or buy AI models according to market demand.

While our proposed framework addresses critical issues in AI model integrity and verification, there are numerous ways for future research and development. By pursuing these directions, we can work towards a more robust, secure, and user-friendly AI model management and distribution ecosystem.

7 Conclusion

As the field of AI continues to evolve rapidly, it is crucial that our supporting infrastructure, including blockchain-based solutions, evolves in tandem to address emerging challenges and opportunities. We have presented a novel framework leveraging Solana's off-chain capabilities for tokenizing AI models, offering significant advancements in traceability and auditability.

With the proposed solution, we have addressed issues in AI model distribution, such as governance, audibility, and security via tokenization on the Solana blockchain. Utilizing blockchain, we have made AI models more tamper-proof and enabled end users to securely perform AI model ownership changes.

We have also shown that published AI models can track their lineage, allowing users to specialize their AI models further and provide concrete evidence of AI model origin and necessary ethical measures.

Further work is also presented on moving to the on-chain implementation of the solution running on the Solana blockchain, improving automation, and allowing advanced security features such as model encryption and access control.

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